

Part 1

Step 7. Setting up a virtual IP address using pfsense

Configuring a pfSense Firewall on the Server

LAB GUIDE

- Before You Begin
- Introduction
- Section 1: Hands-On Demonstration
 - Section 2: Applied Learning
 - Part 1: Configure Network Address Translation

Part 1: Configure Network Address Translation

- Interface: WAN
- Address(es): 10.20.1.3
- Virtual IP Password: password
- Description: WAN VIP for the 172.30.0.10 Network

7. Make a screen capture showing the Virtual IP Address table and paste it into your Lab Report file.

8. From the pfSense menu bar, select Firewall > NAT, then click the 1:1 tab to configure the NAT mapping.

9. On the NAT Mappings page, select Add to bind the new WAN IP address to the internal network.

10. Use the following information to complete the Edit NAT 1:1 Entry form, then Save and Apply Changes.

Unless specified in the following list, accept the default value in the form.

- External subnet: 10.20.1.3
- Internal IP Type: Single host
- Internal IP Address: 172.30.0.10
- Description: NAT 10.20.1.3 to 172.30.0.10

11. Make a screen capture showing the NAT Mappings table and paste it into your Lab Report file.

Virtual IP Address

Virtual IP address	Interface	Type	Description	Actions
10.20.1.3/32 (vhid: 1)	WAN	CARP	WAN VIP for 172.30.0.10	Edit

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Step 11. Setting up NAT to configure connection between WAN and LAN IP addresses

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Part 1: Configure Network Address Translation

- Virtual IP Password: password
- Description: WAN VIP for the 172.30.0.10 Network

7. Make a screen capture showing the Virtual IP Address table and paste it into your Lab Report file.

8. From the pfSense menu bar, select Firewall > NAT, then click the 1:1 tab to configure the NAT mapping.

9. On the NAT Mappings page, select Add to bind the new WAN IP address to the internal network.

10. Use the following information to complete the Edit NAT 1:1 Entry form, then Save and Apply Changes.

Unless specified in the following list, accept the default value in the form.

- External subnet: 10.20.1.3
- Internal IP Type: Single host
- Internal IP Address: 172.30.0.10
- Description: NAT 10.20.1.3 to 172.30.0.10

11. Make a screen capture showing the NAT Mappings table and paste it into your Lab Report file.

NAT 1:1

Port Forward 1:1 Outbound NPT

NAT 1:1 Mappings

Interface	External IP	Internal IP	Destination IP	Description	Actions
☒ WAN	10.20.1.3	172.30.0.10	*	NAT 10.20.1.3 to 172.30.0.10	Add Edit Delete Save

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Part 2

Step 3. Set up of WAN Rules to allow traffic to come into Private network complete.

The screenshot displays the pfSense Firewall configuration interface. The sidebar on the left contains a 'LAB GUIDE' with sections for 'Section 1: Hands-On Demonstration', 'Section 2: Applied Learning', and 'Section 3: Lab Challenge and Analysis'. The main configuration area shows the 'Firewall / Rules / WAN' tab. A green notification banner at the top states: 'The settings have been applied. The firewall rules are now reloading in the background. Monitor the reload progress.' Below this, the 'Rules (Drag to Change Order)' table is visible, listing several rules for the WAN interface.

States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✗ 0/0/0	*	Reserved Not assigned by IANA	*	*	*	*	*		Block bogon networks	⚙️
☐ ✓ 0/0/0	IPv4 TCP	*	*	172.30.0.10	80 (HTTP)	*	none		Web Browsing	⬇️ ⬆️ ⬇️ ⬆️ ⬇️ ⬆️
☐ ✓ 0/0/0	IPv4 TCP	*	*	172.30.0.10	443 (HTTPS)	*	none		Secure Web	⬇️ ⬆️ ⬇️ ⬆️ ⬇️ ⬆️
☐ ✓ 0/0/0	IPv4 TCP	*	*	172.30.0.10	53 (DNS)	*	none		Domain Name Service	⬇️ ⬆️ ⬇️ ⬆️ ⬇️ ⬆️
☐ ✓ 0/0/0	IPv4 TCP	*	*	172.30.0.10	25 (SMTP)	*	none		Mail	⬇️ ⬆️ ⬇️ ⬆️ ⬇️ ⬆️

At the bottom of the interface, a taskbar shows various application icons, including a search bar, file explorer, and web browser. The system tray at the bottom right indicates the time as 10:43 AM on 9/20/2021, with a weather forecast of 78°F and 'Raining now'.

Step 9. Showing that the firewall works and traffic is passing through to be received by TargetWindows01.

The screenshot displays a web browser window with the following elements:

- Browser Tabs:** PG Campus, Course Schedule - IT104 Intro, Bongo, Unit 4 Lab - IT104 Intro, Jones & Bartlett, Lab View.
- Address Bar:** <https://jbl-iti.hatsize.com/labview/?e=1540969>
- Page Title:** Configuring a pfSense Firewall on the Server
- Virtual Machine dropdown menu:**
 - LAB GUIDE
 - Section 1: Hands-On Demonstration
 - Section 2: Applied Learning
 - Part 1: Configure Network Address Translation
 - Part 2: Allow Access from a Public Address
 - Section 3: Lab Challenge and Analysis
 - Part 2: Allow Access from a Public Address
 - Virtual Machine dropdown menu
- Lab Steps:**
 - On the TargetWindows01 machine, launch a Command Prompt window.
 - At the command prompt, execute `route add 172.30.0.0 mask 255.255.255.0 10.20.1.1` to route the incoming traffic.
 - From the TargetWindows01 taskbar, launch Internet Explorer and navigate to <http://10.20.1.3>.
 - Make a screen capture showing the <http://10.20.1.3> IIS default page and paste it into the Lab Report file.
 - On your local computer, save the updated spreadsheet as `yourname_PFSense-FW-PLANNER-NAT-S2.xls`, replacing `yourname` with your own name.
- Note:** This completes Section 2 of this lab. Submit the `yourname_PFSense-FW-PLANNER-NAT-S2.xls` file you just completed as part of your deliverables.

The main content area shows the IIS default page with the title "Internet Information Services" and a grid of "Welcome" messages in various languages. The Windows taskbar at the bottom includes the Start button, search bar, task view, and several pinned application icons. The system tray shows the date and time as 10:51 AM on 9/20/2021, along with weather information (79°F, Raining now).